

VAN-AN HOANG

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EXPERIENCE

Pixta - the largest online stock media marketplace in Asia

Junior/ Intermediate AI Research Engineer

Hanoi, Vietnam

Oct 2023 - Aug 2025

- **AutoReview Project**

- **Engineered** a unified MLOps data pipeline using **ClearML** to integrate AWS S3, DynamoDB, and BigQuery, streamlining data processing for model training.
- **Evaluated and integrated** SOTA Vision-Language Models (including **Qwen2-VL** and **Gemini 2.0**) for trademark recognition; optimized deployment based on latency and inference cost analysis.
- **Developed** an automated image processing algorithm achieving **85% recall (NG)** and **92% recall (Accept)**, resulting in a cost reduction of **\$1,500/month** in manual review expenses.

- **AI Search Project (Semantic Search Engine for Pixtastock)**

- **Architected** a semantic search backend using **OpenSearch** and **Weaviate** (HNSW-PQ indexing) for the publicly launched "AI Search" feature on pixta.jp.
- **Achieved 90% search result acceptance** by implementing a vision-language fusion approach utilizing **RAM++**, **Tag2Text**, and **Osprey** for pixel-level visual understanding.
- **Built** a multi-stage detection pipeline combining **YOLO-World** for open-vocabulary object detection and facial attribute recognition.
- **Deployed** autoscaling, asynchronous inference endpoints on **AWS SageMaker** with a robust CI/CD pipeline for deep learning modules.

- **Independent Research & Technical Writing**

- **Authored** a technical deep-dive on **InstaFlow** (one-step diffusion architecture) published on **Viblo**, Vietnam's largest tech-sharing platform.
- **Researched** state-of-the-art architectures (**Mamba**, **DiffuSSM**) and developed a PoC for image generation using **Flux**.

DSLAb - BKAI International Research Center

Research Assistant

Hanoi, Vietnam

Nov 2022 - July 2024

- Efficient training diffusion models with SNR re-weighting and novel resampling strategy

Project link

- Resampled timestep with novel and more adaptive distribution
- Employed reversed Signal to Noise Ratio in Loss function
- Detailed Empirical Results with an acceleration in convergence speed and better FID, IS

- Researching on Deep Generative Models: Autoregressive Models, Flow Models, Generative Adversarial Network, Diffusion Probabilistic Models

Viettel Aerospace Institute

AI Engineer Intern

Hanoi, Vietnam

May 2023 - Sep 2023

- Redesigned body-face joint model github.com/AibeeDetect/BFJDet with a new post hoc for landmark detection (training code in pytorch) and write the evaluation code for landmark detection (metrics used: Mean Square Error, Root Mean Square Error)
- Ranked 27/165 Viettel Programming Challenge - A nationwide competition of Viettel Group on algorithm programming.
- Researched on Neural Radiance Fields (NeRF) and its Application on View Synthesis

EDUCATION

University of Technology Sydney, Sydney, Australia

Master of Artificial Intelligence

Aug 2025 — Now

Overall GPA: 6.8/7 (rounded)

Achievement and award

UTS Academic Excellence International Scholarship

Hanoi University of Science and Technology, Hanoi, Vietnam

Bachelor degree in Data Science and Artificial Intelligence

Oct 2020 — Oct 2024

Overall GPA: 3.66

Achievement and award

Excellent Scholarship - Type A (top 5%) - GPA: 3.93

The Scholarship Program Sponsored by the Electricity of Vietnam (EVN)

Machine Learning Teaching Assistant

April 2024 — June 2024

PROJECTS

Stokkie - AI-Powered Stock Photography Toolkit

Hanoi, Vietnam

Project link

Oct 2025 – Present

- Built a full-stack AI productivity platform for stock contributors using **React 19**, **TypeScript**, **Firebase**, and **Google Gemini/Imagen APIs** with secure serverless architecture.
- Designed a **node-based Canvas workflow** system for visual image pipelines, supporting AI generation, enhancement, inpainting, batch analysis, and multi-image operations.
- Implemented multi-modal AI features: professional image critique, SEO metadata generation, compliance analysis (model/property release detection, content safety), and text-to-image generation.
- Developed a **privacy-first RAG chatbot** with in-browser vector search and image context support for contributor guidance.

Gaither - Autonomous Recruiting Agent Swarm

Sydney, Australia

Project link

Dec 2025

- Built a multi-agent AI recruiting platform using **Python**, **FastAPI**, **React 19/Next.js 16**, and **Claude AI** with real-time WebSocket streaming for live agent activity visualization.
- Architected a **3-agent pipeline** (Hunter, Analyzer, Engager) that autonomously discovers GitHub candidates, evaluates code quality with LLM-powered scoring, and generates personalized outreach messages.
- Integrated **Weaviate vector database** for semantic candidate search.
- Developed an **AI chatbot with grounding tools** enabling recruiters to query candidate data, compare profiles, analyze GitHub activity, and generate interview questions with real-time web search.

Efficient Training Diffusion Models

Hanoi, Vietnam

Project link

March 2024 - July 2024

- Resampled timestep with novel and more adaptive distribution
- Employed reversed Signal to Noise Ratio in Loss function
- Detailed Empirical Results with an acceleration in convergence speed and better FID, IS

Analysis on different methods of rain removal task

Hanoi, Vietnam

Project link

Oct 2022 - Jan 2023

- Fine-tuned Gated Context Aggregation Network on the task of removing rain drop on lens when capturing landscape and give a benchmark test on SSIM and PSNR to compare with other models such as DDPM, GAN, Transweather

Document Summarization

Hanoi, Vietnam

Project link

Dec 2022 - Jan 2023

- Implement Sequence to Sequence (Seq2Seq) RNN with pytorch
- Fine-tuning model google mt5-small on processed cnn.dailymail dataset and obtain Rouge-1: 28.102, Rouge-2: 13.426, Rouge-L: 24.824 (comparable to GPT-2), outperform implemented Seq2Seq model.

SKILLS

- **English:** IELTS 7.5 Overall
- **Programming:** Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Pytorch Lightning, Matplotlib, Plotly), Java, C++, SQL, Git, ElasticSearch, OpenSearch, Generative AI, ComfyUI, Flux, Stable Diffusion

REFERENCES

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